

MTH204 (Section A)

Quiz-2

April 15, 2026

1 hour and 10 minutes

Total Marks: 30

Name:

Roll No.:

Important Notice: Only a pen and a single **A4-sized handwritten cheat sheet** are permitted during the quiz. No other materials, devices, or aids are allowed. **No discussion or communication of any kind is permitted.** Any violation of these rules will result in the award of **zero marks** for the quiz, and the incident will be formally reported.

Q1.

(10 marks)

Use the power series method to solve the following differential equation:

$$2x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - 3)y = 0.$$

Q2.

(8 marks)

Solve the following I.V.P. using Laplace Transform:

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 5y = h(t),$$

where

$$h(t) = \begin{cases} 1, & 0 < t < \pi, \\ 0, & t > \pi, \end{cases}$$

and

$$y(0) = 0, \quad y'(0) = 0.$$

Q3.

(6 marks)

Find the inverse Laplace transform of the following:

(a) $\mathcal{L}^{-1}\left\{\tan^{-1}\left(\frac{1}{s}\right)\right\}$ (3 marks)

(b) $\mathcal{L}^{-1}\left\{\frac{1}{s}\ln\left(\frac{s+a}{s+b}\right)\right\}$ (3 marks)

Q4.

(6 marks)

Show that

$$\int x^2 J_0(x) dx = x^2 J_1(x) + x J_0(x) - \int J_0(x) dx.$$